Contrasting a Model Against Its Own Interventions: Hop-Graph Contrastive Decoding for Causally Grounded Multi-Hop Reasoning

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Abstract

Language models often reach the right answer without using the reasoning chain they write, and fail when spurious context or parametric shortcuts dominate. Context-aware decoding (CAD) and its variants counter this by contrasting the next-token distribution with and without the context. We give a causal reading of CAD: its contrast estimates the *total* effect of the *whole* context, including paths through distractors that have no causal edge to the answer. We propose HGCD (Hop-Graph Contrastive Decoding), a training-free decoding rule that contrasts the model against *interventions on its own reasoning graph*. HGCD (i) parses the model's chain into a directed reasoning graph (DRG), (ii) intervenes on each answer-relevant hop together with its graph descendants, and (iii) re-weights answers by their probability plus α times the average *total effect* of the chain on them, computed in probability space. We prove that (a) the common "mask one step" estimator measures a controlled direct effect that vanishes on bridge-entity chains, (b) probability-space contrast cancels shortcut and distractor components of a mixture reader exactly, whereas CAD amplifies distractor components, and (c) in the $\alpha \to \infty$ limit HGCD selects the answer with maximal hop-level probability of necessity and sufficiency (PNS). We pair the method with an evaluation protocol in which every intervention is a binary, programmatic manipulation with a *known* ground-truth effect (renaming, reordering, distractor injection, placebo insertion: true effect 0; inserted redundant hops: true necessity 0), on CLUTRR (controlled kinship graphs, ID $k \leq 4$ / OOD $k \geq 5$) and MuSiQue (real text, bridge-entity chains). All code runs end to end on a free Kaggle GPU.

1 Introduction

Chain-of-thought prompting [30] makes models write intermediate steps, but writing a step is not the same as *using* it. Causal tests show that models frequently ignore or post-rationalise their own chains [9, 14, 26, 27], answer multi-hop questions through single-hop shortcuts [2, 12, 32], and are easily derailed by irrelevant context [19]. Two families of remedies exist. *Training* remedies (counterfactual preference objectives [8, 14], PNS-filtered supervision [33], causality-aware post-training [6]) are effective but costly. *Decoding* remedies are training-free: contrastive decoding [11, 13], classifier-free guidance [18], and context-aware decoding (CAD) [20] with its adaptive and gated successors [23, 29].

CAD scores tokens by $(1+\alpha)\log \pi(\cdot \mid c, x) - \alpha \log \pi(\cdot \mid x)$. Read causally, the log-ratio is the effect of $\mathrm{do}(C{=}c)$ versus $\mathrm{do}(C{=}\varnothing)$: the total effect of the *entire* context. In multi-hop QA the context contains supporting paragraphs on the causal path to the answer and distractors that are not; CAD amplifies both. This explains, mechanistically, the empirical failures of CAD on uninformative or distracting context that motivated AdaCAD and NWCAD [23, 29].

Idea. Instead of contrasting *with versus without context*, contrast the model's answer distribution *given its own chain* against the distribution under *interventions on the hops of that chain*, keeping the context fixed. Because the context is held fixed in both terms, distractor and parametric components cancel; because interventions follow the chain's own graph, the contrast measures the *path-specific* contribution of the reasoning steps.

Contributions.

1. **A causal account of contrastive decoding** for multi-hop reasoning: CAD estimates a total context effect that includes non-causal distractor paths (proposition 3).
2. **HGCD**, a training-free rule that contrasts the model against graph-aware interventions on its own reasoning chain, with an optional evidence-level variant (L1) that uses a *type-matched counterfactual context* instead of an empty one (section 4).
3. **Theory**: naive step masking measures a controlled direct effect that is zero on bridge chains (proposition 1); probability-space contrast cancels shortcut and distractor mixture components exactly (proposition 2); HGCD converges to a hop-level PNS rule (proposition 4), connecting decoding to the identifiability results of Yu et al. [33].
4. **A verifiable evaluation protocol**: binary, programmatic interventions with known ground-truth effects, an estimator-validation benchmark with known necessity labels, placebo controls, and paired statistics; everything reproducible on one free T4 GPU.

2 Related Work

Faithfulness of reasoning chains. Lanham et al. [9] intervene on model-written chains (truncation, inserted mistakes, paraphrases); Turpin et al. [26] show biased features steer answers without being verbalised; FRODO [14] uses causal mediation [28] and trains small models with counterfactual preferences; FUR [27] measures faithfulness by unlearning steps; Thought Anchors [3] measures step importance by counterfactual resampling. We use interventions not only to *measure* but to *decode*.

Necessity and sufficiency. Yu et al. [33] define step-level probabilities of necessity and sufficiency for chain-of-thought, prove identifiability under exogeneity and monotonicity (building on Pearl [16], Tian and Pearl [24]), and use PNS to prune chains for in-context learning and fine-tuning. We show the same quantity arises as the limit of a decoding rule, and we separate direct from total effects [15].

Contrastive and guided decoding. Contrastive decoding [11, 13], DoLa [4], CFG [18], CAD and its adaptive/gated variants [20, 23, 29] contrast distributions under different models, layers or contexts. Visual contrastive decoding [10] uses a *distorted* rather than empty input; our L1 variant is the textual analogue with a type-matched counterfactual. Pced [5] treats each retrieved document as an expert contrasted with a no-context baseline; RED [31] multiplies image- and rationale-conditional distributions in vision-language models. None of these intervene on the hops of the model's own reasoning graph, distinguish direct from total effects, or relate the contrast to PNS.

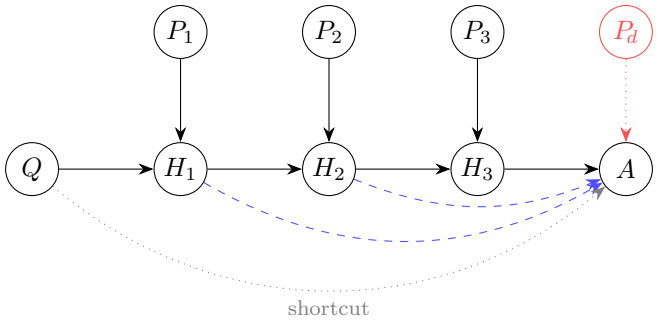


Figure 1: Task SCM. Solid: bridge chain. Dashed blue: extra edges of compositional chains. Red dotted: spurious distractor path that the *reader* may use but the task graph forbids. Gray: parametric shortcut. CAD's contrast includes the red path; HGCD holds P fixed and contrasts interventions on H_j only.

Causal analyses of reasoning. Stolfo et al. [22] quantify robustness of mathematical reasoning through interventions on a causal graph of the input; Zečević et al. [34] argue models "talk" causality without being causal; text-intervention effect estimation [7] uses LLM rewrites as treatments. Our interventions are programmatic, so their ground-truth effects are known exactly.

3 Setup: a Structural Causal Model of Multi-Hop Reasoning

Variables. A question Q, a context $P = (P_{\text{sup}}, P_{\text{dist}})$ of supporting and distractor passages, a chain of hops $H_1, \ldots, H_k$, and an answer A. Each hop is rendered as `Hop` i`:` ⟨`body`⟩ `=>` ⟨`result`⟩. The model acts as a *reader* mechanism $\pi(a \mid Q, P, h_{1:k})$ over a finite candidate set $\mathcal{A}$.

Two task graphs. In *bridge* chains (MuSiQue), hop j resolves an entity that the next hop consumes and the answer is the result of the last hop: $Q \to H_1 \to \cdots \to H_k \to A$, with $P_{\text{sup},j} \to H_j$. In *compositional* chains (CLUTRR), every hop contributes a relation that is composed into the answer: $H_j \to A$ for all j and $H_j \to H_{j+1}$ through the shared entity. Distractors have no edge to A in either task graph (fig. 1).

Directed reasoning graph (DRG). Given a generated chain $y = (h_1, \ldots, h_k)$ we add an edge $i \to j$ $(i < j)$ if the result of h_i is mentioned in the body of h_j. Terminal hops are those whose result is the query target (CLUTRR) or the last hop (MuSiQue); the *relevant set* $\mathcal{R}(y)$ contains the terminal hops and their DRG ancestors. Hops outside $\mathcal{R}(y)$ have no directed path to the answer.

Interventions on hops.

Definition 1 (Naive and graph-aware interventions)**.** y_j^{nai} *replaces hop* j *by a mask token and leaves all other hops unchanged.* y_j^{tot} *masks hop* j *and, in every DRG descendant* d *of* j*, masks every mention of the results of* j *and of* d*. The latter is a deterministic, rollout-free approximation of* $\text{do}(H_j{=}\perp)$*: descendant values are functions of* h_j*, hence unknown under the intervention. A* placebo *rewrites hop* j *without changing its entities or relation.*

For an answer a we write $\Delta_j^{\text{nai}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{nai}})$ and $\Delta_j^{\text{tot}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{tot}})$, suppressing Q, P.

Algorithm 1 HGCD (L2) for one question

$y \leftarrow$ greedy chain from the model (optionally with L1 contrast on result tokens)
Parse hops; build DRG; compute relevant set $\mathcal{R}(y)$
$p \leftarrow \pi(\cdot \mid Q, P, y)$ over $\mathcal{A}$ ▷ one forward + KV-cached candidates
for $j \in \mathcal{R}(y)$ **do**
    $p_j \leftarrow \pi(\cdot \mid Q, P, y_j^{\text{tot}})$
end for
return $\arg\max_{a \in \mathcal{V}_\beta} (1+\alpha) p(a) - \alpha \operatorname{mean}_j p_j(a)$

4 Method: Hop-Graph Contrastive Decoding

Answer-level rule (L2). With uniform weights $w_j = 1/|\mathcal{R}(y)|$ on relevant hops and a plausibility set $\mathcal{V}_\beta = \{a : \pi(a \mid y) \geq \beta \max_{a'} \pi(a' \mid y)\}$ [11],

$$\hat{a} = \arg\max_{a \in \mathcal{V}_\beta} s_\alpha(a), \qquad s_\alpha(a) = (1+\alpha)\,\pi(a \mid y) - \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \pi(a \mid y_j^{\text{tot}}) = \pi(a \mid y) + \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \Delta_j^{\text{tot}}(a). \tag{1}$$

The score is the answer probability plus α times the average total effect of the answer-relevant hops on that answer. Redundant hops ($j \notin \mathcal{R}(y)$) are never contrasted.

Evidence-level rule (L1). While emitting the *result* tokens of a hop, we contrast the real context P with a type-matched counterfactual $\tilde{P}$ of identical length and entity types in which the relational content differs (CLUTRR: names permuted within gender; MuSiQue: supporting passages swapped for distractors): $t^\star = \arg\max_{t \in \mathcal{V}_\beta} (1+\alpha)\pi(t \mid P, \cdot) - \alpha\pi(t \mid \tilde{P}, \cdot)$. Unlike an empty context, $\tilde{P}$ is in-distribution and controls for length and type cues.

Cost. L2 needs $1 + |\mathcal{R}(y)|$ prefix forwards and a KV-cached pass over the candidate tokens; no extra generation. L1 doubles the forward cost of result tokens only.

Choice of α. α is selected on the CLUTRR validation split and frozen; an adaptive variant sets $\alpha \cdot \mathrm{JSD}(p, \bar{p}_j)$ as in AdaCAD [29].

5 Theory

All proofs are short and given in appendix A; the propositions make predictions that our protocol tests directly.

Assumption 1 (Markov reader on bridge chains)**.** *For bridge chains, $\pi(a \mid h_{1:k})$ depends on $h_{1:j}$ only through $h_{j+1:k}$ whenever h_{j+1} restates the result of h_j ($j < k$).*

Proposition 1 (Naive masking is a direct effect)**.** $\Delta_j^{\text{nai}}(a)$ *equals the controlled direct effect of H_j on A with $H_{j+1:k}$ held at their observed values. Under assumption 1, $\Delta_j^{\text{nai}}(a) = 0$ for every $j < k$ on bridge chains, whereas $\Delta_j^{\text{tot}}(a)$ is not constrained to zero. On compositional chains, where A reads each hop's relation, the direct effect is generally non-zero.*

Prediction. Naive masking separates necessary from redundant hops on CLUTRR but not on non-final MuSiQue hops; graph-aware masking works on both (table 1).

Assumption 2 (Mixture reader)**.** $\pi(a \mid Q, P, y) = \lambda_c \pi_c(a \mid y, P_{\text{sup}}) + \lambda_s \pi_s(a \mid Q) + \lambda_d \pi_d(a \mid P_{\text{dist}})$*, with* λ*'s non-negative, summing to one, and invariant to interventions on hops.*

Proposition 2 (Exact cancellation in probability space)**.** *Under assumption 2,* $s_\alpha(a) = \pi(a \mid y) + \alpha\lambda_c \sum_j w_j [\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})]$*: the shortcut* π_s *and distractor* π_d *components cancel in the contrast for every* α*. The log-space analogue does not cancel them.*

Proposition 3 (CAD amplifies distractor paths)**.** *Under assumption 2 applied to direct answering, with no-context distribution* $\pi_s(a \mid Q)$*, CAD's score is* $(1+\alpha)\log\left[\lambda_c\pi_c + \lambda_s\pi_s + \lambda_d\pi_d\right](a) - \alpha \log \pi_s(a \mid Q)$*. For any* $\alpha > 0$ *and any gold answer* g*, a distractor answer* a_d *with* $\pi_d(a_d) = 1$ *is preferred by CAD whenever* $\pi_s(a_d \mid Q)$ *is small enough, even if* $\pi(g \mid Q, P) > \pi(a_d \mid Q, P)$*. Adding or removing* P_{dist} *changes CAD's decision but leaves HGCD's contrast term unchanged.*

Prediction. Under distractor injection, CAD's accuracy degrades more than HGCD's, i.e. $|\text{ATE}_{\text{distract}}|$ is larger for CAD.

Proposition 4 (PNS limit)**.** *Interpret* $\pi(a \mid y_j^{\text{tot}})$ *as* $P(A{=}a \mid \text{do}(H_j{=}\perp))$ *and assume exogeneity and monotonicity as in Yu et al. [33]. Then* $\Delta_j^{\text{tot}}(a) = \text{PNS}_j(a)$ *and* $\lim_{\alpha\to\infty} s_\alpha(a)/\alpha = \sum_{j\in\mathcal{R}(y)} w_j \,\text{PNS}_j(a)$*. Hence, within* $\mathcal{V}_\beta$*, HGCD interpolates between the most probable answer (*$\alpha{=}0$*) and the answer for which the relevant hops are, on average, most necessary and sufficient.*

6 Experimental Protocol

Data. CLUTRR [21] (`gen_train234_test2to10`), restricted to clean simple-path stories (the filter is reported), gives exact gold paths; ID $k \in \{2, 3, 4\}$, OOD $k \geq 5$; answers are scored over the 20 kinship relations. **MuSiQue-Ans** [25] gives real text with gold decompositions; the context holds the supporting paragraphs plus 4 distractors. MuSiQue was built to resist single-hop shortcuts [12], which is why we prefer it to HotpotQA. Candidates for MuSiQue are the model's own answer, its hop results and the paragraph titles (never the gold answer); we report candidate recall.

Binary interventions with known effects.

Intervention	Definition (programmatic, seeded)	True effect on A
`rename`	all names → fresh fictional names (gender-preserving)	0
`order`	shuffle story sentences / paragraphs	0
`distract` / `dist8`	insert a disconnected family / 4 extra distractor paragraphs	0
`placebo_input`	insert two neutral sentences without entities	0
redundant hop	insert a true but off-path fact as a hop of a gold chain	necessity 0

For each method we report the paired ATE $\mathbb{E}[Y(1) - Y(0)]$ on accuracy with bootstrap 95% CIs; a robust method has ATE ≈ 0. `placebo_input` controls for "any change hurts".

Estimator validation. Gold chains with one inserted redundant hop give hop-level labels (necessary = 1, redundant = 0). We report AUC of Δ^{tot}, Δ^{nai}, the placebo Δ, rollout PNS [33] and the DRG path indicator, overall and on chains the model answers correctly (PS = 1), and mean effects by hop position.

Methods. Direct answering; CoT; CAD [20]; AdaCAD [29]; CAD-chain (contrast with vs. without the chain, a text analogue of rationale-conditioned decoding); HGCD and ablations: naive masking, no graph (all hops), log space, adaptive α, placebo contrast (must equal CoT); L1 chains decoded with CoT or HGCD.

Pre-registered hypotheses.

Hypothesis 1 (Estimator)**.** *AUC*(Δ^{tot}) ≥ 0.75 *on both datasets; AUC*(Δ^{nai}) *near chance on MuSiQue non-final hops; placebo* $\Delta \approx 0$*.*

Hypothesis 2 (Accuracy)**.** *HGCD > CoT and > CAD-chain on OOD CLUTRR (*$k \geq 5$*) and on MuSiQue, with paired bootstrap CIs excluding zero.*

Hypothesis 3 (Robustness)**.** $|\text{ATE}_{\text{distract}}|$ *of HGCD < that of CAD (proposition 3); ATE of* `placebo_input` ≈ 0 *for all methods.*

Hypothesis 4 (Ablations)**.** *HGCD* $\geq$ *naive-mask and no-graph variants; probability space* $\geq$ *log space; placebo contrast = CoT.*

Models and compute. Qwen2.5-1.5B-Instruct [17] (main) and SmolLM2-1.7B-Instruct [1] (second family) on two Kaggle T4 GPUs in fp16; Qwen2.5-7B-Instruct on one A100 for the scale point. Greedy decoding; $\beta = 0.1$; $\alpha \in \{0, 0.25, 0.5, 1, 2, 4\}$ chosen on the validation split.

Statistics. Bootstrap CIs (2,000 resamples), paired bootstrap differences and exact McNemar tests on the same questions.

7 Results

All tables in this section are generated by the notebook; nothing is typed by hand.

Table 1: Estimator validation (AUC for separating necessary from inserted redundant hops).

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

Table 2: CLUTRR accuracy (%) with bootstrap 95% CIs. ID: $k \leq 4$; OOD: $k \geq 5$.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

Table 3: CLUTRR paired ATE (accuracy points) of binary interventions whose true effect is 0.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

8 Limitations

Assumptions. ?? 1**??** 2 idealise the reader; we treat the propositions as testable predictions rather than guarantees, and the protocol is designed to falsify them. **Rollout-free interventions.** y_j^{tot} approximates do(H_j) by masking rather than re-generating descendants; we validate it against

Table 4: MuSiQue exact match (%) and paired ATE.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

rollout PNS where compute allows. **Format dependence.** The DRG requires hop-structured chains; we report format success and fall back to CoT when parsing fails. **Candidate sets.** Answer-level contrast needs a finite candidate set; it is exact on CLUTRR and approximate on MuSiQue (we report candidate recall). **Scale.** Main experiments use 1.5–7B models; larger models may rely less on shortcuts. **Data filter.** We use CLUTRR stories whose gold graph is a simple path; stories with back-edges contain known label ambiguities.

9 Conclusion

Contrasting a model against interventions on its own reasoning graph turns step-level causal measurement into a training-free decoding rule. The rule is cheap, comes with falsifiable predictions, and is evaluated with interventions whose true effects are known.

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A Proofs

Proposition 1. y_j^{nai} changes only h_j and keeps $h_{j+1:k}$ at their observed values, so Δ_j^{nai} is by definition the controlled direct effect $\pi(a \mid h_j, h_{-j}) - \pi(a \mid \perp, h_{-j})$ [15]. Under assumption 1, $\pi(a \mid h_{1:k}) = g(a \mid h_{j+1:k})$ for $j < k$, so both terms equal $g(a \mid h_{j+1:k})$ and the difference is zero. y_j^{tot} additionally changes $h_{j+1:k}$ (their results are masked), so g is evaluated at a different argument and the difference is unconstrained. On compositional chains π reads the relation of every hop, so removing h_j alone changes the composed relation. □

Proposition 2. Substituting assumption 2 into eq. (1), with $\sum_j w_j = 1$ and π_s, π_d unaffected by interventions on hops: $s_\alpha(a) = (1+\alpha)[\lambda_c \pi_c(a \mid y) + \lambda_s \pi_s + \lambda_d \pi_d] - \alpha \sum_j w_j [\lambda_c \pi_c(a \mid y_j^{\text{tot}}) + \lambda_s \pi_s + \lambda_d \pi_d] = \pi(a \mid y) + \alpha \lambda_c \sum_j w_j [\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})]$. In log space, $\log(\lambda_c u + r) - \log(\lambda_c v + r)$ depends on $r = \lambda_s \pi_s + \lambda_d \pi_d$ unless $u = v$, so the shortcut terms do not cancel. □

Proposition 3. Let $m(a) = \pi(a \mid Q, P)$. CAD prefers a_d over g iff $(1+\alpha)\log\frac{m(a_d)}{m(g)} > \alpha \log\frac{\pi_s(a_d)}{\pi_s(g)}$. The left side is bounded below by $(1+\alpha)\log\frac{\lambda_d}{m(g)}$ since $m(a_d) \geq \lambda_d$, while the right side tends to $-\infty$ as $\pi_s(a_d) \to 0$. For HGCD, both terms of eq. (1) condition on the same P, so $\lambda_d \pi_d$ enters both and cancels (proposition 2). □

Proposition 4. Under exogeneity and monotonicity, Lemma 2 of Yu et al. [33] gives PNS = $P(A{=}a \mid \mathrm{do}(S)) - P(A{=}a \mid \mathrm{do}(S'))$. With $S = y$ and $S' = y_j^{\mathrm{tot}}$ this is $\Delta_j^{\mathrm{tot}}(a)$. Dividing eq. (1) by α and letting $\alpha \to \infty$ removes the $\pi(a \mid y)/\alpha$ term. □

B Implementation and Reproducibility

Candidate log-probabilities are computed with one prefix forward and a KV-cached pass over the candidate tokens (verified against full recomputation to 2×10^{-5}). Prompts are two-shot, hop-structured; the second CLUTRR shot uses a relation outside the label set to avoid leaking label statistics. All interventions are seeded functions of the example id. The Kaggle notebook embeds the full package and writes every table in this paper.

C Pilot run (pipeline sanity check)

The following tables come from a CPU smoke run with Qwen2.5-0.5B-Instruct on a handful of examples. They verify that the pipeline runs end to end and are **not** evidence for or against any hypothesis.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

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